

ORE Intraday Power Curves and Derivatives

30 April 2026

Document History

Date	Author	Comment
April 2026	LSEG	initial version

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1 Introduction

Commodity curves in ORE return a single price per delivery date. This is sufficient for products that settle against a daily price, but power markets trade at a finer granularity. Intraday products reference prices for specific parts of the day, e.g. the 09:00–10:00 hour on a weekday.

The earlier approach modelled this with dedicated load curves, e.g. peak and off-peak curves built from the respective futures. A separate curve was needed for every delivery pattern.

The intraday power framework replaces these dedicated curves with a single curve per underlying that carries an intraday shape, combined with a trade-specific load profile. This supports flexible delivery windows without a new curve per pattern. The scope of this first version is limited to intraday power forwards and commodity swaps with intraday power floating legs; exposure simulation of these products is not yet covered.

A day-level price comes from a commodity price curve. Intraday shape factors, provided as market data, scale this price across the buckets of the day. A trade specifies which buckets it delivers into through its load profile. The pricing of overlapping intervals and the treatment of daylight saving are described in Section 2.1.

2 Pricing

2.1 Intraday Power Pricing

Intraday power pricing converts a daily-average commodity price into a load-weighted average price per MWh by applying intraday shape factors and a delivery-period load profile.

2.1.1 Components

Intraday power pricing combines three main components:

- **Commodity price curve:** Provides the underlying daily-average power prices as a function of delivery date. These are typically constructed from futures prices using backward-flat interpolation.
- **Intraday shape factors:** Define the relative price profile across hours of the day. Shape factors partition the day into buckets (e.g., peak hours 08:00–20:00 and off-peak hours 20:00–08:00 the next day or even more granular) and assign a multiplicative weight to each bucket.
- **Load profile:** Specifies the trades the energy load (in MW) for each time interval during the delivery period. For example, 50 MW load from 07:15 to 07:30.

2.1.2 Daily Intraday Price

For a given delivery date d , the daily-average price P_d^{day} from the commodity curve is scaled by a load-weighted average shape factor to give the intraday load weighted price:

$$P_d^{\text{intraday}} = P_d^{\text{day}} \times \bar{s}_d^{\text{load}}.$$

For each load bucket i (with constant load ℓ_i MW over $[\text{start}_i, \text{end}_i]$), two quantities are computed:

- The energy delivered: $M_{d,i} = \ell_i \cdot \Delta t_{d,i} / 3600$ (MWh, where $\Delta t_{d,i}$ is the DST-adjusted duration in seconds),
- The time-weighted shape factor: $s_i = s(\text{start}_i, \text{end}_i)$ (see Section below).

The load-weighted average shape factor is then:

$$\bar{s}_d^{\text{load}} = \frac{\sum_{i=1}^n M_{d,i} \cdot s_i}{\sum_{i=1}^n M_{d,i}},$$

i.e. each shape factor s_i is weighted by the energy $\ell_i \cdot \Delta t_{d,i}$ delivered in that bucket.

2.1.3 Time-Weighted Shape Factor

Shape factors are typically given as a discrete step function: a constant multiplier w_j applies over each interval $[t_j, t_{j+1})$. The time-weighted average shape factor over $[\text{start}, \text{end}]$ is:

$$s(\text{start}, \text{end}) = \frac{\sum_j w_j \cdot \max(0, \min(\text{end}, t_{j+1}) - \max(\text{start}, t_j))}{\text{end} - \text{start}}.$$

This formula iterates over all shape intervals and computes the overlap of each interval $[t_j, t_{j+1})$ with $[\text{start}, \text{end}]$, weighted by the shape factor w_j on that interval.

2.1.4 Load Profile and Energy Calculation

A load profile specifies a constant load ℓ_i (in MW) for each intraday time interval $[\text{start}_i, \text{end}_i]$. On delivery date d , the applicable profile is determined by forward-flat filling: the profile with the latest effective date $\leq d$ is used. Two representations are supported: explicit profiles keyed by date, or rule-based profiles with separate shapes for business days and non-business days.

On a normal day the duration is simply $\Delta t_{d,i} = \text{end}_i - \text{start}_i$ seconds. DST transitions adjust this:

- **Forward-shift date** (spring forward, 23-hour day): the hour 02:00–03:00 does not exist. Any bucket overlapping that window has its duration reduced by the overlap; buckets entirely within 02:00–03:00 contribute zero energy and are excluded from the weighted average.
- **Backward-shift date** (fall back, 25-hour day): an extra hour 02:00–03:00 is repeated. Load segments marked with the `dst` flag represent the load in that extra hour and are included only on backward-shift dates. The use either specific weights for the DST hour or fall back to use the regular scaling.

2.1.5 Period Average Price

Over a calculation period from start date to end date, the average intraday price is the daily-load-weighted average of the daily intraday prices:

$$P_{\text{period}} = \frac{\sum_{d \in \mathcal{D}} M_d^{\text{total}} \cdot P_d^{\text{intraday}}}{\sum_{d \in \mathcal{D}} M_d^{\text{total}}},$$

where:

- $\mathcal{D}$ is the set of pricing dates in the period (determined by the pricing calendar and business-day flags),
- $M_d^{\text{total}} = \sum_i M_{d,i}$ is the total energy on day d (after DST adjustment),
- P_d^{intraday} is the load-weighted intraday price on day d .

The period amount is then:

$$\text{Amount} = Q \cdot P_{\text{period}} \cdot G + S,$$

where Q is the quantity, G is the gearing multiplier, and S is the additive spread. The quantity interpretation depends on the **QuantityMode**:

- **TotalEnergy**: Q is the total MWh for the full period; $\sum_d M_d^{\text{total}}$ is normalized to determine the relative delivery across days.
- **LoadShapeMultiplier**: Q is a multiplier applied to the absolute total load; $\sum_d M_d^{\text{total}}$ scales the fixed period energy.

2.1.6 Example

Consider a simple example with:

- Delivery period: single day, 2026-06-25 (normal day, 24 hours),
- Commodity price: $P_d^{\text{day}} = 100$ USD/MWh,
- Load profile: constant 50 MW from 01:00 to 08:00 (7 hours),
- Shape factors: 0.8 from 01:00 to 05:00, 1. from 05:00 to 08:00.

First, compute the energy in each segment:

- Segment 1 (01:00–05:00): $M_1 = 50 \text{ MW} \times 4 \text{ h} = 200 \text{ MWh}$,
- Segment 2 (05:00–08:00): $M_2 = 50 \text{ MW} \times 3 \text{ h} = 150 \text{ MWh}$,
- Total: $M_d^{\text{total}} = 350 \text{ MWh}$.

Next, compute the load-weighted average shape factor:

$$\bar{s}_d^{\text{load}} = \frac{200 \times 0.8 + 150 \times 1.5}{350} = \frac{160 + 225}{350} = \frac{385}{350} = 1.1$$

Finally, compute the intraday price:

$$P_d^{\text{intraday}} = 100 \times 1.1 = 110.0 \text{ USD/MWh.}$$

If the period consists of this single day with `QuantityMode = TotalEnergy` and $Q = 700$ MWh, the period amount is:

$$\text{Amount} = 700 \times 110. = 77000 \text{ USD.}$$

2.1.7 Backward-Flat Filling

Both the commodity price curve and the shape factors are constructed using backward-flat interpolation on the delivery date axis. This means that:

- If the commodity curve provides monthly prices with backward-flat interpolation, all days in a month reference that month's price until a new monthly price takes effect.
- Similarly, if shape factors are updated on specific dates, the latest shape profile applies to all subsequent days until a new profile is provided.

This sparse schedule approach allows for compact representation of seasonal and calendar patterns.

2.1.8 Daylight-Saving Adjustment

On a daylight-saving forward-shift date (spring forward, e.g., USA transitions at 02:00 to 03:00), the day has 23 hours. The missing hour between 02:00 and 03:00 is excluded from both the total energy calculation and the shape factor weighting. Any load segments that span 02:00–03:00 have their duration reduced accordingly.

On a daylight-saving backward-shift date (fall back, e.g., USA transitions at 02:00 back to 01:00), the day has 25 hours. An extra hour exists between 02:00 and 03:00. Load segments marked with the `dst` flag (intended to specify the load for this extra hour) contribute to the calculation. Load segments without the flag are applied to the entire day, including the repeated hour, unless explicit logic excludes them.

See Section 4.0.1 for details on the load profile structure and the `dst` attribute.

3 Curve Configuration and Market Data

An intraday power curve has two components. A commodity price curve provides the day-level price, and a set of intraday shape factors distributes this price across the buckets of the day. The shape factors are market data and are loaded through the curve configuration.

3.1 Curve Configuration

3.1.1 Intraday Power Curves

An intraday power curve augments a daily-average commodity price curve with intraday shape factors. The daily-average curve provides the day-level price and the shape factors distribute this price across intraday buckets.

Note: if the underlying commodity curve uses **BackwardFlat** interpolation and the market quotes are monthly averages (rather than daily quotes), the intraday shape factors can also be used to model seasonality within the month.

Listing 1 shows an example configuration.

```
<IntradayPowerCurves>
  <IntradayPowerCurve>
    <CurveId>PJM_WH_RT_PK_HOURLY</CurveId>
    <CurveDescription>PJM Western Hub Real Time Hourly</CurveDescription>
    <Currency>USD</Currency>
    <DailyAveragePriceCurve>Commodity/USD/PJM_WH_RT_PK</DailyAveragePriceCurve>
    <ShapeQuoteName>PJM_WH_RT_PK_HOURLY</ShapeQuoteName>
    <Convention>PJM_WH_RT_PK</Convention>
  </IntradayPowerCurve>
</IntradayPowerCurves>
```

Listing 1: Intraday power curve configuration

The meaning of the elements is:

- **CurveId**: Unique identifier of the intraday power curve.
- **CurveDescription** [Optional]: Free text description.
- **Currency**: Curve currency.
- **DailyAveragePriceCurve**: Full curve spec of the underlying daily-average commodity curve, e.g. `Commodity/USD/PJM_WH_RT_PK`.
- **ShapeQuoteName**: Prefix used to load intraday shape factor quotes. The quote pattern is `SHAPE_PROFILE/SHAPE_FACTOR/<ShapeQuoteName>/*` where the wildcard expands to the shape key, e.g. `<ProfileDate>/<TimeUnitsFromStartOfDay>/<Unit>/<DST_FLAG>`. See [Section 3.2](#) for the full market datum format.
- **Convention**: Commodity future convention id used determine the region for daylight-saving handling in the intraday shape term structure.

Shape profiles are applied with flat forward fill on the delivery date axis. If shape quotes are present for Monday and then next on Saturday of the same week, Monday's shape is used for Monday to Friday and Saturday's shape is used from Saturday onward.

This allows sparse shape schedules. If the shape is constant, one profile date is enough. If the shape is constant within each month, profiles can be provided only at month start dates. If the shape changes by weekday block, profiles can be provided on representative days, e.g. Monday and Saturday.

3.2 Intraday Power Shape Factor

Property	Allowable values	Description
Instrument Type	<i>SHAPE_PROFILE</i>	Intraday power shape profile quote.
Quote Type	<i>SHAPE_FACTOR</i>	Multiplicative intraday shape factor.
Quote Name	String	Shape profile identifier, e.g. market region/profile id.
Delivery Date	Date string	Date for which the intraday profile is defined.
Start Time	Integer	Start offset from beginning of day in the given time unit.
Unit	<i>SEC, HOUR</i>	Time unit used for the start time offset.
DST Flag [Optional]	<i>DST</i>	Marks a factor used for daylight-saving handling.

Table 1: Intraday Power Shape Factor

The quote format is:

- `SHAPE_PROFILE/SHAPE_FACTOR/<QuoteName>/<DeliveryDate>/<StartTime>/<Unit>[/DST]`

Examples:

- `SHAPE_PROFILE/SHAPE_FACTOR/PJM_WH_RT_PK_HOURLY/2021-01-01/0/SEC`
- `SHAPE_PROFILE/SHAPE_FACTOR/PJM_WH_RT_PK_HOURLY/2021-01-01/3600/SEC`
- `SHAPE_PROFILE/SHAPE_FACTOR/PJM_WH_RT_PK_HOURLY/2021-01-01/7200/SEC`
- `SHAPE_PROFILE/SHAPE_FACTOR/PJM_WH_RT_PK_HOURLY/2021-03-01/7200/SEC/DST`

The quote value is always a price in currency per MWh during the delivery window.

Assume for illustration that these four quotes represent prices in USD/MWh for each delivery window. The first quote is for 00:00–01:00, the second for 01:00–02:00 and the third for 02:00 onward according to the profile definition. The last quote with the optional DST flag is only used on a backward DST day for the extra 02:00–03:00 hour.

For backward DST handling, there must be a DST quote starting at 02:00 and all DST quotes for the day must have a start time strictly before 03:00.

Worked examples:

- Assume the daily average price is 100 USD/MWh, the shape factor for `SHAPE_PROFILE/SHAPE_FACTOR/PJM_WH_RT_PK_HOURLY/2021-01-01/3600/SEC` is 1.10 and the contract delivers 40 MW from 01:15 to 01:30. The delivered energy is

$$40 \times \frac{15}{60} = 10 \text{ MWh}$$

and the delivery falls fully into the 01:00–02:00 bucket, so the price is

$$100 \times 1.10 = 110 \text{ USD/MWh.}$$

- Assume the daily average price is again 100 USD/MWh, the 01:00–02:00 shape factor is 1.10 and the 02:00 onward shape factor from `SHAPE_PROFILE/SHAPE_FACTOR/PJM_WH_RT_PK_HOURLY/2021-01-01/7200/SEC` is 1.00. If the contract delivers 50 MW from 01:30 to 02:30, the first half hour contributes 25 MWh in the 01:00–02:00 bucket and the second half hour contributes 25 MWh in the 02:00–03:00 bucket. The weighted price is

$$100 \times \frac{25 \times 1.10 + 25 \times 1.0}{50} = 105 \text{ USD/MWh.}$$

The shape resolution is flexible. A single curve can mix buckets of different lengths, e.g. 15 minute buckets for the today and the next day, hourly buckets further out and daily buckets beyond that. Alternatively, separate curves can be set up per granularity for the same underlying, e.g. a 15 minute curve and an hourly curve. Separate curves are reported on different risk factor keys, so their stress and sensitivity results needs to be aggregated or simulated through curve algebra (see the curve algebra section of the User Guide).

4 Reusable Load Profiles

A load profile defines intraday time buckets and the load in for each delivery bucket. We can specify the load profile on a trade level or reuse a profile defined in the conventions.

4.0.1 Intraday Power Load Conventions

A node with name `IntradayPowerLoad` is used to store reusable intraday power load profiles. These conventions can be referenced from `PowerLoadProfileReference` in an `IntradayPowerFloatingLegData` node. The same `PowerLoadProfileData` structure can also be given inline in the leg itself.

Listing 2: Intraday power load conventions

```
<IntradayPowerLoad>
  <Id>...</Id>
  <PowerLoadProfileData>
    ...
  </PowerLoadProfileData>
</IntradayPowerLoad>
```

The meanings of the elements in this node are as follows:

- **Id:** Unique identifier of the load profile convention.
- **PowerLoadProfileData:** Load profile definition used to convert daily-average prices into load-weighted intraday prices.

`PowerLoadProfileData` supports two mutually exclusive representations:

- **ExplicitDates:** Explicit load profiles by effective date. Profiles are forward-flat filled, i.e. if a delivery date has no profile, the latest profile with effective date less than or equal to the delivery date is used.

- **BusinessDayRules:** Rule-based load profiles with a calendar and separate business-day and non-business-day profiles. Rules are also forward-flat filled by effective date.

In both cases, load is given by one or more **LoadFactor** elements. Each **LoadFactor** has required attributes **from** and **to**, an optional **unit** attribute and an optional **dst** flag. If **unit** is omitted, seconds are assumed. The element value is the load in MW over the interval. The **dst** flag is intended for the extra hour between 02:00 and 03:00 on a daylight-saving backward-shift date.

Listing 3: Power load profile data examples

```

<PowerLoadProfileData>
  <ExplicitDates>
    <LoadProfileDatum>
      <Date>2026-06-25</Date>
      <LoadFactors>
        <LoadFactor from="1" to="24" unit='HOUR'>1.0</LoadFactor>
      </LoadFactors>
    </LoadProfileDatum>
  </ExplicitDates>
</PowerLoadProfileData>

<PowerLoadProfileData>
  <BusinessDayRules>
    <LoadProfileBusinessDayRule>
      <Date>2026-01-01</Date>
      <Calendar>US-NERC</Calendar>
      <BusinessDayLoadFactors>
        <LoadFactor from="8" to="20" unit='HOUR'>1.0</LoadFactor>
      </BusinessDayLoadFactors>
      <NonBusinessDayLoadFactors>
        <LoadFactor from="1" to="24" unit='HOUR'>0.6</LoadFactor>
      </NonBusinessDayLoadFactors>
    </LoadProfileBusinessDayRule>
  </BusinessDayRules>
</PowerLoadProfileData>

```

In the first example, the profile is given explicitly for the date 2026-06-25. It defines a constant load of 1 MW from hour 1 to hour 24 on that date. If no later explicit profile is provided, this profile is also used for later delivery dates until it is replaced by another explicit dated profile.

In the second example, the profile is rule-based from 2026-01-01 onward. The **Calendar** determines whether a delivery date is treated as business day or non-business day. On business days in **US-NERC**, the active load profile is 1 MW from 08:00 to 20:00. On non-business days, the active load profile is 0.6 MW from 01:00 to 24:00. As with explicit profiles, the rule is forward-flat filled and remains in force until a later rule date is provided.

Pricing uses the load profile to convert a daily-average price into a load-weighted daily price per MWh.

On a daylight-saving forward date, the missing hour between 02:00 and 03:00 is excluded from both the delivered MWh and the load-weighted shape average. For example, assume a load profile of 50 MW from 01:00 to 08:00 and a daily-average price

of 100 USD/MWh. On a normal day, the delivered energy is $50 \times 7 = 350$ MWh. On a forward day, the 02:00–03:00 hour does not exist, so the delivered energy becomes

$$50 \times 6 = 300 \text{ MWh.}$$

5 Historical Fixings

The load profile of a trade determines the granularity of the historical fixings. An hourly profile expects hourly fixings, a 15 minute profile expects 15 minute fixings.

Fixing names are derived from the underlying name, the delivery date and the delivery window. The day-level name has the form

`POWER-<UnderlyingName>-<DeliveryDate>`, where the delivery date is an ISO date, e.g. `POWER-PJM_WH_RT_PK-2021-06-01`. For an intraday bucket the start and end of the window in seconds from midnight are appended, e.g.

`POWER-PJM_WH_RT_PK-2021-06-01-32400-36000` for the 09:00–10:00 hour ($9 \times 3600 = 32400$, $10 \times 3600 = 36000$). The extra hour created when clocks move back at the end of daylight saving carries an additional `-DST` suffix. A bucket that covers the full day (0 to 86400 seconds) uses the plain day-level name.

For a trade with a load profile, one fixing is expected per bucket with positive load. The index fixing for the delivery date is the load-weighted average of the bucket fixings. Without a load profile, a single full-day fixing under the day-level name is used.

Intraday fixings on the delivery date itself are supported. On the evaluation date, buckets whose fixing has already been published are taken from the history, and buckets without a published fixing are forecast from the curve. For an hourly profile delivering from 09:00 to 13:00, observed at 11:00 with fixings published for the 09:00 and 10:00 buckets, those two fixings are used and the 11:00 and 12:00 buckets are forecast. For dates strictly before the evaluation date all expected bucket fixings must be present.

6 Supported Trades

The intraday power framework currently supports forwards and swaps.

6.1 Forwards

6.1.1 Intraday Power Forward

An Intraday Power Forward contract is an agreement between two counterparties to buy/sell power for a specified intraday delivery window, at a predetermined price (the strike). An intraday power forward does not involve any upfront payment.

The `IntradayPowerForwardData` node is the trade data container for the `IntradayPowerForward` trade type. The structure of example `IntradayPowerForwardData` nodes is shown in Listings [4](#) and [5](#).

Listing 4: Intraday power forward data with explicit delivery window

```
<IntradayPowerForwardData>
  <Position>Short</Position>
  <Maturity>2021-05-27</Maturity>
  <Name>PJM_WH_RT_PK</Name>
  <Currency>USD</Currency>
  <Strike>32.9</Strike>
  <Quantity>1</Quantity>
  <DeliveryDate>2021-06-01</DeliveryDate>
  <DeliveryStart>900</DeliveryStart>
  <DeliveryEnd>1800</DeliveryEnd>
  <IsDstHour>false</IsDstHour>
  <PhysicallySettled>false</PhysicallySettled>
  <PaymentDate>2021-06-03</PaymentDate>
</IntradayPowerForwardData>
```

Listing 5: IntradayPowerForwardData with PowerLoadProfileData

```
<IntradayPowerForwardData>
  <Position>Long</Position>
  <Maturity>2026-06-30</Maturity>
  <Name>PJM_WH_RT_PK</Name>
  <Currency>USD</Currency>
  <Strike>40.25</Strike>
  <Quantity>2</Quantity>
  <DeliveryDate>2026-07-01</DeliveryDate>
  <PowerLoadProfileData>
    <BusinessDayRules>
      <LoadProfileBusinessDayRule>
        <Date>2026-01-01</Date>
        <Calendar>US-NERC</Calendar>
        <BusinessDayLoadFactors>
          <LoadFactor from="8" to="20" unit="HOUR">1.0</LoadFactor>
        </BusinessDayLoadFactors>
      </LoadProfileBusinessDayRule>
    </BusinessDayRules>
  </PowerLoadProfileData>
</IntradayPowerForwardData>
```

The meanings and allowable values of the elements in the `IntradayPowerForwardData` node follow below.

- **Position:** Defines whether the underlying power will be bought (long) or sold (short).
Allowable values: *Long, Short*
- **Maturity:** The maturity date of the forward contract.
Allowable values: Any date string, see **Date** in Table ??.
- **Name:** The name of the underlying intraday power index or curve.
Allowable values: See **Name** for commodity trades in Table ??.
- **Currency:** The currency the underlying power is quoted in. The strike and forward price are both considered to be in this currency.
Allowable values: See **Currency** in Table ??.

- **Strike**: The agreed buy/sell price of the intraday power forward.
Allowable values: Any real number.
- **Quantity**: The number of total energy (in MWh) bought/sold.
Allowable values: Any real number.
- **DeliveryDate**: Delivery date for the intraday power forward.
Allowable values: Any date string, see **Date** in Table ??.
- **DeliveryStart** [Optional]: Start of delivery window, expressed in seconds from start of day on **DeliveryDate**. Must be provided together with **DeliveryEnd** when **PowerLoadProfileData** is not provided.
Allowable values: Any integer.
- **DeliveryEnd** [Optional]: End of delivery window, expressed in seconds from start of day on **DeliveryDate**. Must be provided together with **DeliveryStart** when **PowerLoadProfileData** is not provided.
Allowable values: Any integer.
- **IsDstHour** [Optional]: Flag to mark the repeated hour on backward-shift dates when using explicit delivery window representation. If omitted, it defaults to **false**.
Allowable values: Any string that evaluates to true or false as outlined in Table ??.
- **PowerLoadProfileData** [Optional]: Alternative to **DeliveryStart** and **DeliveryEnd**. This defines delivery load by profile. One must provide either both **DeliveryStart** and **DeliveryEnd**, or **PowerLoadProfileData**.
Allowable values: See Section 4.0.1.
- **PhysicallySettled** [Optional]: A value of **true** indicates physical settlement. A value of **false** indicates cash settlement. If omitted, it defaults to *true*.
Allowable values: Any string that evaluates to true or false as outlined in Table ??.
- **PaymentDate** [Optional]: If **PhysicallySettled** is set to *false*, this gives the cash settlement date. It must be greater than or equal to the **Maturity** date. If omitted and the forward is cash settled, the **Maturity** date is used.
Allowable values: Any date string, see **Date** in Table ??.
- **SettlementData** [Optional]: This node is used to specify the settlement of the cash flows for cash settled forwards, and the payment flow for physically settled ones.

A **SettlementData** node is shown in Listing 6, and the meanings and allowable values of its elements follow below.

- **PayCurrency**: The settlement currency for the payment cashflow.
Allowable values: See **Currency** in Table ??.
- **FXIndex**: The FX reference index for determining the FX fixing at the value date. The forward price is observed at maturity and the NPV is converted to **PayCurrency** with the **FXIndex** using an FX fixing on **FixingDate** (settlement date), discounted from **PaymentDate**.

Allowable values: The format of the **FXIndex** is “FX-FixingSource-CCY1-CCY2” as described in Table ??.

- **FixingDate**: The date on which the **FXIndex** is observed.
Allowable values: See **Date** in Table ??.

Listing 6: Example SettlementData node

```
<SettlementData>
  <PayCurrency>EUR</PayCurrency>
  <FXIndex>FX-ECB-EUR-USD</FXIndex>
  <FixingDate>2026-07-02</FixingDate>
</SettlementData>
```

6.2 Swaps

Intraday power legs are used within the general commodity swap, see the commodity swap description in the ORE Product Catalogue. The leg uses an intraday power floating leg in place of a standard commodity floating leg.

6.2.1 Intraday Power Floating Leg

An intraday power floating leg is specified in a **LegData** node with **LegType** set to **IntradayPowerFloating**. It is used to define a sequence of cashflows linked to an intraday power index over each calculation period. Pricing is done in two aggregation steps: for each pricing date, a daily price per MWh is computed using the given intraday load shape; daily prices are then averaged over pricing dates in the period using daily volume weights. The period amount is obtained by applying quantity, gearing and spread. The quantity interpretation depends on **QuantityMode**: with *LoadShapeMultiplier*, the load shape defines the absolute intraday and daily loads and **Quantity** scales that absolute shape; with *TotalEnergy*, the load shape is normalized and used as relative weights so that **Quantity** is the total period energy.

For a detailed explanation of the pricing methodology, including the three key components (commodity curve, shape factors, and load profile), formulas for daily and period-average prices, and daylight-saving adjustments, see the intraday power pricing section.

The outline of an intraday power floating leg is given in listing 7. It has the usual **LegData** elements described in section ?? and an **IntradayPowerFloatingLegData** node that is described in section 6.2.2 below.

6.2.2 Intraday Power Floating Leg Data

The **IntradayPowerFloatingLegData** node outline is shown in listing 8. The meaning and allowable values for each node are as follows:

- **Name**: Identifier specifying the intraday power index referenced by the leg.
- **Quantities**: This node specifies constant or period-varying quantities. Usage is analogous to **Notionals** in section ??.

Listing 7: Intraday power floating leg outline.

```
<LegData>
  <LegType>IntradayPowerFloating</LegType>
  <Payer>...</Payer>
  <Currency>...</Currency>
  <PaymentConvention>...</PaymentConvention>
  <PaymentLag>...</PaymentLag>
  <PaymentCalendar>...</PaymentCalendar>
  <ScheduleData>
    ...
  </ScheduleData>
  <PaymentDates>
    <PaymentDate>...</PaymentDate>
  </PaymentDates>
  <IntradayPowerFloatingLegData>
    ...
  </IntradayPowerFloatingLegData>
</LegData>
```

- **Spreads** [Optional]: Optional additive spread applied to the reference price in each calculation period.
- **Gearings** [Optional]: Optional multiplicative gearing applied to the reference price in each calculation period.
- **PricingCalendar** [Optional]: Calendar used to generate pricing dates inside each calculation period.
- **IncludePeriodStart** [Optional]: If *true*, include the period start date in pricing dates.

Allowable values: *true*, *false*. Defaults to *true* if omitted.

- **IncludePeriodEnd** [Optional]: If *true*, include the period end date in pricing dates.

Allowable values: *true*, *false*. Defaults to *false* if omitted.

- **BusinessDays** [Optional]: If *true*, pricing dates are business days in **PricingCalendar**. If *false*, pricing dates are non-business days. Use **NullCalendar** with business days set to *true* for all calendar days.

Allowable values: *true*, *false*. Defaults to *true* if omitted.

- **PowerLoadProfileReference** [Optional]: Reference to an intraday power load profile convention. If no profile is given, it assumes a constant load over 24h.
- **PowerLoadProfileData** [Optional]: Inline load profile data. If both **PowerLoadProfileReference** and **PowerLoadProfileData** are provided, the reference takes precedence. If neither is given, it assumes a constant load over 24h everyday. The structure and semantics of **PowerLoadProfileData** are described in the intraday power load profile conventions section.
- **FXIndex** [Optional]: FX index used to convert each daily fixing before averaging.

- **AvgPricePrecision** [Optional]: Non-negative integer precision for rounding the averaged price.
- **Tag** [Optional]: Optional identifier used to link this leg with other related legs.
- **QuantityMode** [Optional]: Quantity interpretation mode.
 - *TotalEnergy*: The **Quantity** is interpreted as the total MWh for the full calculation period. There are relative load factors derived from the given load profile. The total MWh are delivered relatively to the load profile.
 - *LoadShapeMultiplier*: The **Quantity** is interpreted as a multiplier applied to the period load implied by **PowerLoadProfileData** (or **PowerLoadProfileReference**). In this mode, a load profile is required.

Allowable values: *TotalEnergy*, *LoadShapeMultiplier*. Defaults to *TotalEnergy* if omitted.

Listing 8: Intraday power floating leg data outline.

```

<IntradayPowerFloatingLegData>
  <Name>...</Name>
  <Quantities>
    <Quantity>...</Quantity>
  </Quantities>
  <Spreads>
    <Spread>...</Spread>
  </Spreads>
  <Gearings>
    <Gearing>...</Gearing>
  </Gearings>
  <PricingCalendar>...</PricingCalendar>
  <IncludePeriodStart>...</IncludePeriodStart>
  <IncludePeriodEnd>...</IncludePeriodEnd>
  <BusinessDays>...</BusinessDays>
  <PowerLoadProfileReference>...</PowerLoadProfileReference>
  <PowerLoadProfileData>
    ...
  </PowerLoadProfileData>
  <FXIndex>...</FXIndex>
  <AvgPricePrecision>...</AvgPricePrecision>
  <Tag>...</Tag>
  <QuantityMode>...</QuantityMode>
</IntradayPowerFloatingLegData>

```

7 Parameterisation

7.1 Simulation Market

Intraday power curves in the simulation market are configured in an **IntradayPowerCurves** node under **Market**. The node structure is:

- **Simulate** [Optional]: If present, enables or disables simulation of intraday power curves. If omitted, it defaults to **false**.
- **Names**: List of intraday power curve names to include.

- **Tenors**: Pillar tenors used to build simulated curve points. Tenors can be provided per name using attribute **name**, or as a default tenor set with empty **name** (or no attribute).
- **Interpolation** [Optional]: Interpolation method for each intraday power curve. Can be specified per name via attribute **name** or as a default. If omitted, the default interpolation is **Linear**.

Simulation currently evolves the day-average prices of intraday power price curve only. Intraday shape factors are kept constant during simulation and are used to distribute simulated day-average prices to intraday buckets.

If **UseSpreadedTermStructures** is enabled in the sensitivity setup, **LogLinear** and **LogLinearFlat** interpolation should not be used for intraday power curves, because spreaded values can become negative.

At least one **Name** and at least one **Tenors** node are required. If tenors are not provided for each named curve explicitly, a default tenor set must be provided.

An example configuration is shown in Listing 9.

Listing 9: Simulation market intraday power curves configuration

```
<IntradayPowerCurves>
  <Simulate>true</Simulate>
  <Names>
    <Name>PJM_WH_RT_PK</Name>
  </Names>
  <Tenors name="">1W,2W,1M,2M,3M,6M,1Y</Tenors>
  <Interpolation name="">Linear</Interpolation>
</IntradayPowerCurves>
```

Curve linking and algebraic mapping for intraday power curves are described in Section ??.

It can be preferable to simulate intraday power curves via their underlying commodity curves and report sensitivities on the commodity curve keys, for example during sensitivity analysis or stresstesting.

This can be achieved configured using **CurveAlgebra** as shown below.

An intraday power mapping example is shown in Listing 10.

7.2 Sensitivity

Intraday power sensitivity configuration is provided in an **IntradayPowerCurves** block under **SensitivityAnalysis**. The structure is:

- **IntradayPowerCurves**: Container for intraday power curve sensitivity settings.
- **IntradayPowerCurve** (attribute **name**): Configuration for one intraday power curve.
- **ShiftType**: Shift type, **Absolute** or **Relative**.

Listing 10: CurveAlgebra node for intraday power to commodity mapping

```
<CurveAlgebra>
  <Curve>
    <Key>IntradayPowerCurve/PJM_WH_RT_PK</Key>
    <Operation>
      <Type>Spreaded</Type>
      <Arguments>
        <Argument>CommodityCurve/PJM_WH_RT_PK,1</Argument>
      </Arguments>
    </Operation>
  </Curve>
</CurveAlgebra>
```

- **ShiftSize**: Shift size.
- **ShiftScheme** [Optional]: Finite-difference scheme. If omitted, defaults to Forward.
- **ShiftTenors**: Comma-separated list of tenors to shift.

At present, shocks are applied to the day-average intraday power curve level only. There is no separate parameterisation for shifting individual intraday time buckets directly.

Bucket-level shocked prices are implied from the day-level shock and the shape profile:

- For **Relative** shifts, the day-level price is shifted by $n\%$, and bucket prices are shifted by the same relative amount $n\%$.
- For **Absolute** shifts, the day-level price is shifted by an absolute amount, and the implied bucket absolute shift is scaled by the corresponding intraday shape factor.

An example configuration is shown in listing 11.

```
<IntradayPowerCurves>
  <IntradayPowerCurve name="PJM_WH_RT">
    <ShiftType>Absolute</ShiftType>
    <ShiftSize>0.0001</ShiftSize>
    <ShiftTenors>1W,2W,1M,2M,3M,6M,1Y</ShiftTenors>
  </IntradayPowerCurve>
</IntradayPowerCurves>
```

Listing 11: Intraday power curve sensitivity configuration

If sensitivities should be reported on the underlying commodity curve only (instead of separately on intraday power curves), this can be configured in simulation using CurveAlgebra, see Section ?? and Listing ??.

This is useful for mixed portfolios containing both commodity-curve trades and intraday-power-curve trades, where external aggregation of sensitivities is not desired.

The corresponding intraday power mapping example is shown in Listing 10.

In this setup, **Simulate** is set to **true** for **IntradayPowerCurves** in **simulation.xml**, and no **IntradayPowerCurves** shifts are defined in **sensitivity.xml**.

7.3 Stress Testing

7.3.1 Intraday Power Curve Stress Configuration

Intraday power curve stresses are configured in an `IntradayPowerCurves` block under each `StressTest` node.

The structure is:

- `IntradayPowerCurves`: Container for intraday power curve stress definitions.
- `IntradayPowerCurve` (attribute `name`): Stress definition for one intraday power curve.
- `ShiftType`: Shift type, e.g. `Absolute`, `Relative`, or `EqualTo`.
- `Shifts`: Vector of shift values.
- `ShiftTenors`: Vector of tenors corresponding to `Shifts`.

The number of entries in `Shifts` must match the number of entries in `ShiftTenors`.

An example is shown in Listing 12.

Listing 12: Intraday power curve stress configuration

```
<IntradayPowerCurves>
  <IntradayPowerCurve name="PJM_WH_RT_PK">
    <ShiftType>Absolute</ShiftType>
    <Shifts>0.01,0.01,0.01,0.01,0.01,0.01,0.01</Shifts>
    <ShiftTenors>1W,2W,1M,2M,3M,6M,1Y</ShiftTenors>
  </IntradayPowerCurve>
</IntradayPowerCurves>
```

As in sensitivity and simulation, stress is applied to the day-average intraday power price curve. Intraday shape factors are not stressed separately.

Intraday power stress can be represented through the underlying commodity curve via `CurveAlgebra` in simulation configuration, see Section ?? and Listing 10. In that setup, dedicated intraday power curve stress shifts can be omitted.